more than your calculator can explain the numbers you input the last time you used it.

Puzzling over the properties of these particles is not an academic question of why this or that esoteric detail happens to be one way or another. Over the last century, scientists have realized that the universe has the familiar features of common experience only because the particles in Tables 12.1 and 12.2 have precisely the properties they do. Even fairly minor changes to the masses or electric charges of some of the particles would, for example, make them unable to engage in the nuclear processes that power stars. And without stars, the universe would be a completely different place. Thus, the detailed features of the elementary particles are entwined with what many view as the deepest question in all of science: *Why do the elementary particles have just the right properties to allow nuclear processes to happen, stars to light up, planets to form around stars, and on at least one such planet, life to exist?*

The standard model can't offer any insight into this question since the particle properties are part of its required input. The theory won't start to chug along and produce results until the particle properties are specified. But string theory is different. In string theory, particle properties are *determined*by string vibrational patterns and so the theory holds the promise of providing an explanation.

Particle Properties in String Theory

To understand string theory's new explanatory framework, we need to have a better feel for how string vibrations produce particle properties, so let's consider the simplest property of a particle, its mass.

From $E = mc^2$, we know that mass and energy are interchangeable; like dollars and euros, they are convertible currencies (but unlike monetary currencies, they have a fixed exchange rate, given by the speed of light times itself, c^2). Our survival depends on Einstein's equation, since the sun's life-sustaining heat and light are generated by the conversion of 4.3 million tons of matter into energy every second; one day, nuclear reactors on earth may emulate the sun by safely harnessing